Cornell Duffield Engineering
School of Electrical and Computer Engineering

Polar Codes for Higher-Order Alphabets

Author: Jaden Zou (jz2396@cornell.edu) Advisor: Professor Aaron Wagner

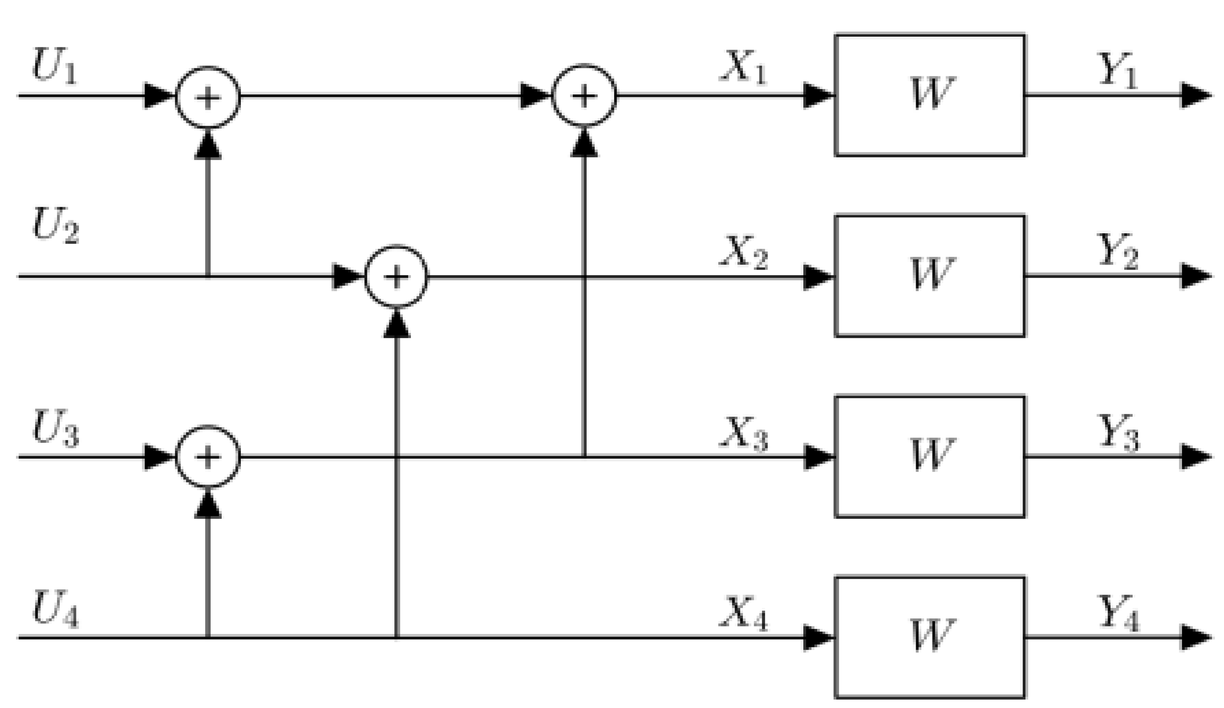
BACKGROUND AND ENCODING

Polar codes (Arıkan, 2009) are the first provably capacity-achieving codes with O(N log N) encoding and decoding complexity. Channel polarization transforms N copies of a channel W into synthetic channels that become either nearly perfect or nearly useless as N grows.

The construction begins with the 2×2 kernel matrix:

$$G_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

The encoder computes $X = U \cdot G_2^{\otimes n}$. For the N=4, $G_2^{\otimes 2}$ will be used to produce the butterfly structure below, mapping input U to codeword X through recursive modular additions (⊕).



BASE IMPLEMENTATION

First implementations of the polar code were over binary discrete memoryless channels (B-DMCs) to validate the framework against known results.

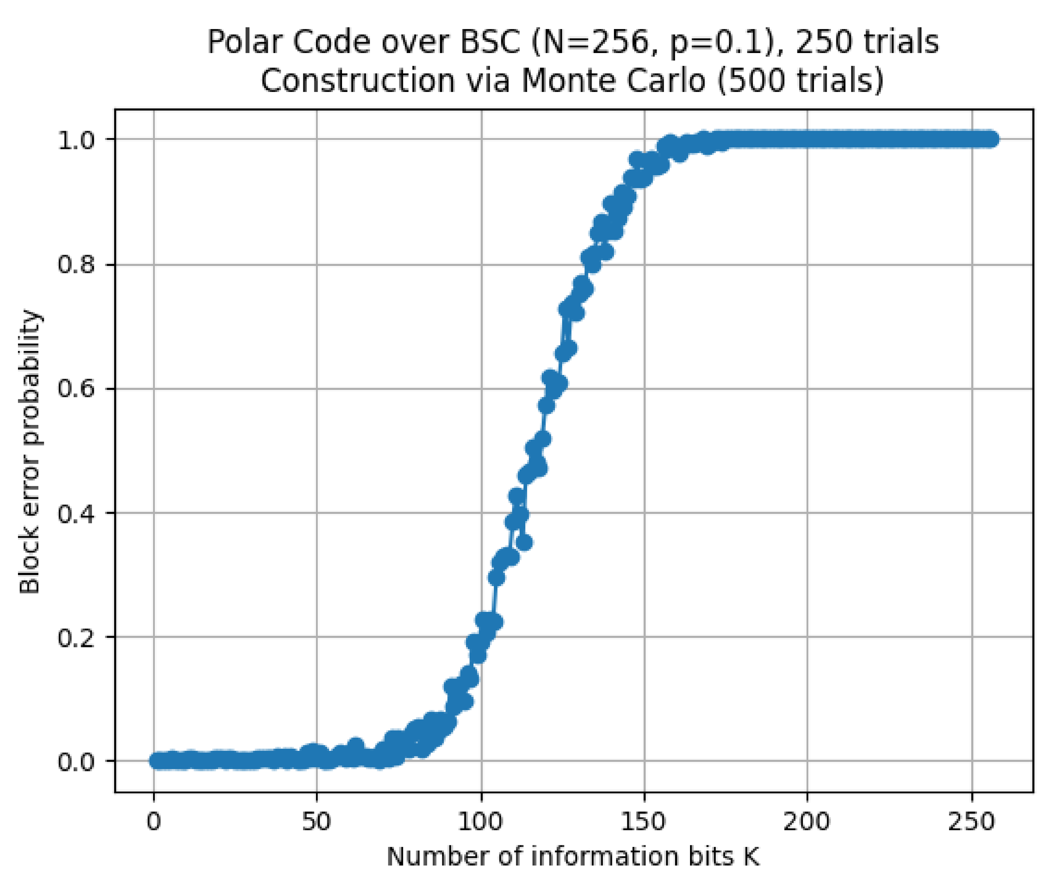
BEC (Binary Erasure Channel): Each input bit is either received correctly or erased with probability ε.

BSC (Binary Symmetric Channel): Each input bit is flipped with crossover probability p.

With N = 256 and p = 0.1, the BSC capacity is:

$$C = 1 - H(p) \approx 0.53 \;\Rightarrow\; K_{max} \approx 136 \text{ bits}$$

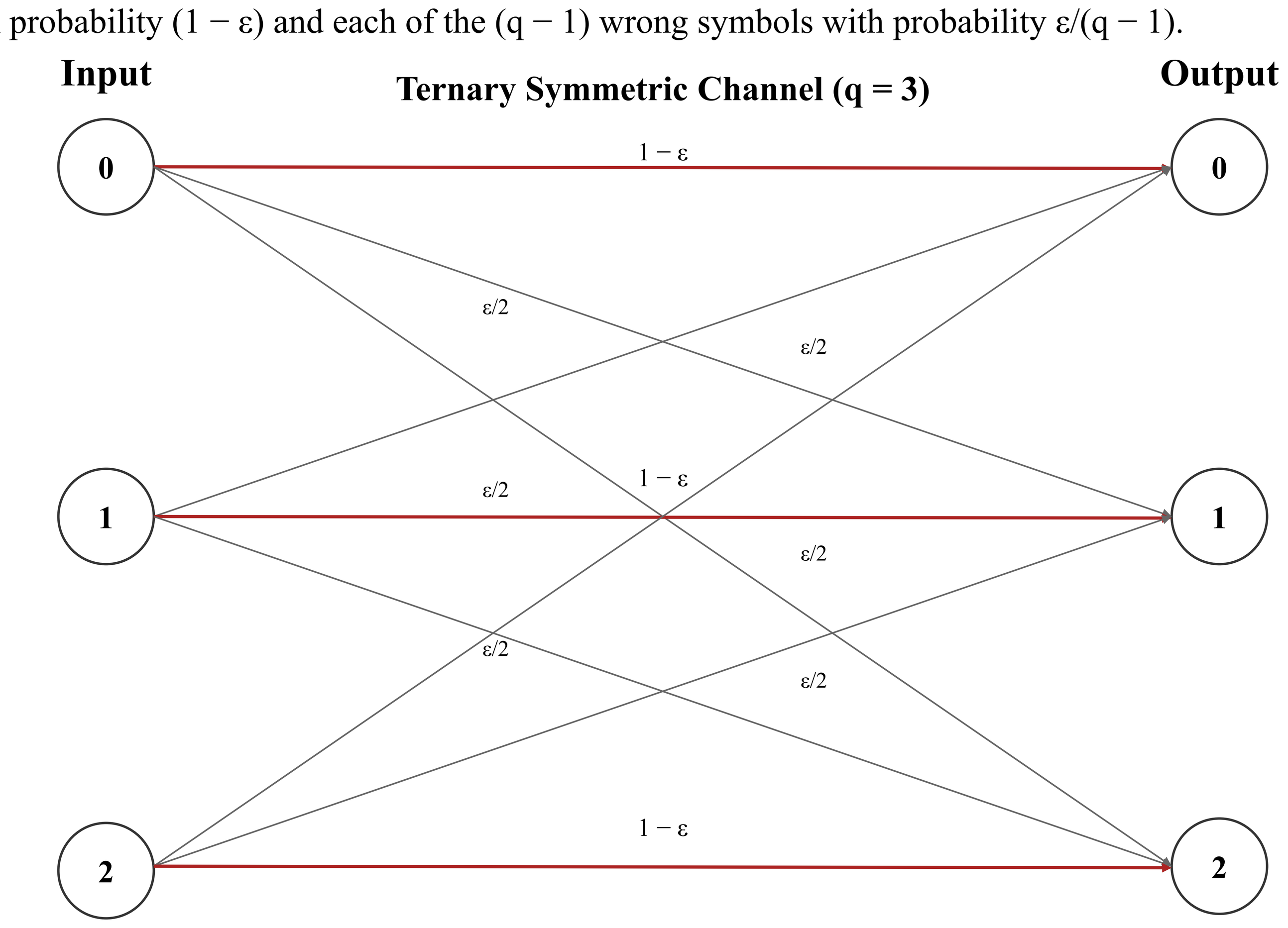
where $H(p) = -p\log_2 p - (1-p)\log_2(1-p)$ is binary entropy



MOVING TO LARGER ALPHABETS

Change of Channel

When the alphabet size q exceeds 2, the Binary Symmetric Channel no longer applies. The natural progression to fix this is the **q-ary Symmetric Channel (q-SC)**, where the correct symbol is received with probability (1 − ε) and each of the (q − 1) wrong symbols with probability ε/(q − 1).



Generalizing Successive Cancellation Decoding to q-ary Channels

Arıkan's original decoding algorithm used likelihood ratios to compare the probabilities of the two possible binary symbol hypotheses. Representing channel beliefs using a single ratio significantly reduced decoding complexity and enabled efficient recursive SC decoding.

Arıkan's Original likelihood ratio:

$$L = \frac{P(y|0)}{P(y|1)}$$

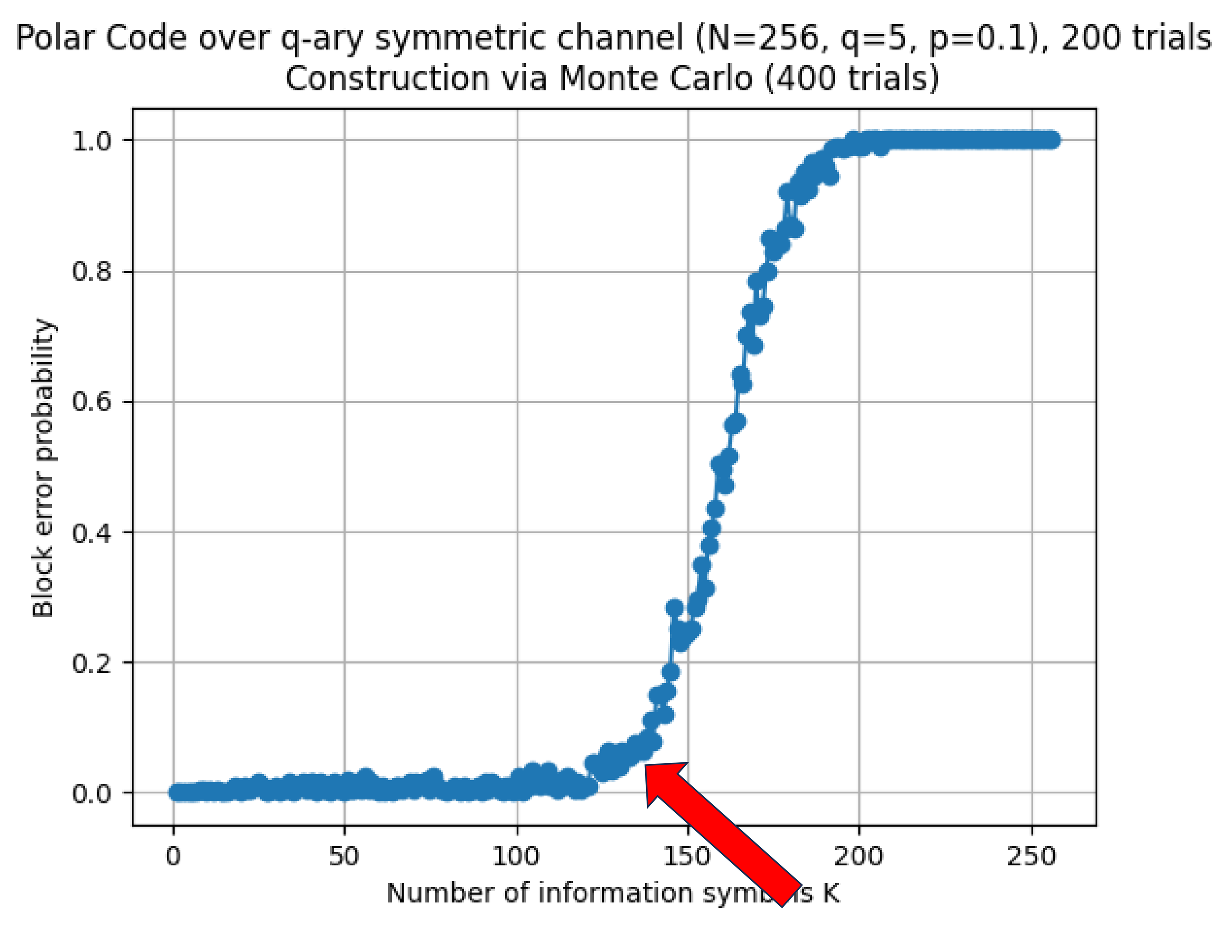
For q-ary channels, multiple symbol candidates must be considered simultaneously, so a single likelihood ratio is no longer. The decoder therefore propagates full symbol probability distributions instead of binary ratios, extending the SC decoding process to larger alphabets.

Decoder complexity increase:

$$O(N\log N) \quad O(q^2 N\log N)$$

INITIAL OBSERVATION

Now, the performance of different alphabet sizes can be compared. Below is an example with q=5.



Achieves higher data rate than q=2

NORMALIZED COMPARISON

There's a problem with the initial comparison: Although good for quick eye-ball comparisons, block error probability graphs penalize larger alphabets unfairly; a q-ary symbol carries the information of $\log_2(q)$ bits, so a single symbol error corrupts more information.

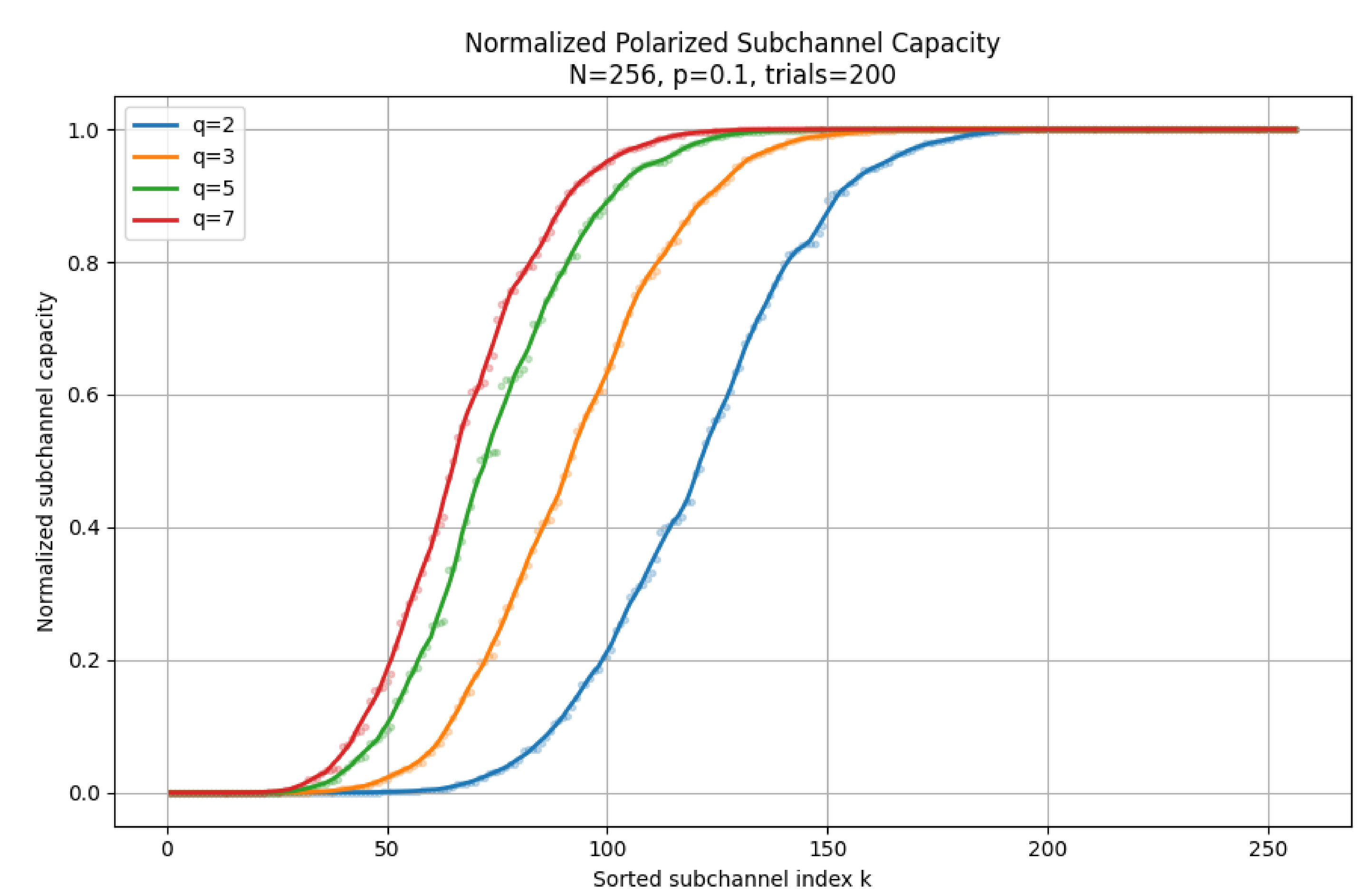
Instead, a fairer comparison metric was used to compare different alphabet sizes by normalizing each synthetic subchannel's capacity to [0, 1] regardless of alphabet size:

$$\hat{C}_k = I(W_k) / \log_2(q) \quad \in [0, 1]$$

where $I(W_k)$ is the mutual information of the k-th synthetic subchannel and $\hat{C}_k$ is the normalized capacity of that subchannel, scaled to [0, 1] for fair comparison across alphabet sizes

The y-axis plots this normalized capacity: subchannels sorted by increasing $\hat{C}_k$ show how polarization drives channels toward 0 (useless) or 1 (perfect), enabling fair comparison across all q values.

The graph below compares normalized subchannel capacity curves for q = 2, 3, 5, 7 under a symmetric channel with N = 256 and p = 0.1.



The graph shows that larger alphabet sizes produce sharper normalized polarization curves, with fewer subchannels remaining in the partially polarized transition region. As q increases, the transition region narrows and shifts left.

This result should be interpreted as finite-length numerical evidence rather than a general proof; the trend may depend on the chosen channel model, flip probability, normalization method, and block length selected.

FURTHER RESEARCH

Several extensions of this work remain open for further investigation:

- Polarization behaviour at different block lengths
- Lower complexity algorithms for decoding q-ary channels
- Analyze why larger alphabets produce sharper normalized polarization curves
- More methods of normalized comparison
- Alternative channel models
- Larger non-prime alphabet sizes

ACKNOWLEDGEMENTS & REFERENCES

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References:
[1] Arıkan, "Channel Polarization," IEEE Trans. IT, 2009.